Comparison and Relationship between Percentage, Percentile Rank, and Percentile

1. Definitions and Formulas

Percentage

A **percentage** expresses a value as a proportion of a whole, standardized to 100:

$$Percentage = \left(\frac{\text{Part}}{\text{Whole}}\right) \times 100$$

Percentile Rank

The **percentile rank** of a score indicates the percentage of values in a dataset that are less than or equal to that score:

$$Percentile\ Rank = \left(\frac{L + 0.5E}{N}\right) \times 10$$

Where:

- L = number of values less than the score
- E = number of values equal to the score
- N = total number of values

Percentile (Value)

A **percentile** indicates the **actual data value** below which a given percentage kkk of the data falls.

To estimate the **k-th percentile** in a sorted dataset:

$$P_k = x_{\left(\frac{k}{100} \times (n+1)\right)}$$

Where:

- P_k = the k-th percentile value
- x = the ordered dataset
- n = total number of values
- k = the desired percentile (with $k \in [0{,}100]$), representing the percentage of data expected to fall **below** the corresponding value.

Note: Interpolation is typically used when the index is not an integer.

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2. Key Differences

Criterion	Percentage	Percentile Rank	Percentile (Value)
Nature	Absolute measure	Relative position in a dataset	Actual value corresponding to a rank position
Unit	% of total	% of values below or equal	Data value (not a percentage)
Dependency	Independent of dataset	Dependent on data distribution	Dependent on data distribution
Range	0–100	0–100	Determined by scale of data
Requirement	Individual value and total	Full dataset	Full dataset
Use Case	Reporting scores or rates	Ranking individuals within a population	Setting thresholds or cut-off values

3. Mathematical Relationship

There is no direct conversion between **percentage** and **percentile** without knowledge of the **data distribution**.

- A **percentage** represents an individual's score in isolation.
- A **percentile rank** depends on how that score compares with others in the group.
- A **percentile (value)** refers to the actual score that separates a given percentage of the dataset.

In a **normal distribution**, the *relationship between percentage scores and percentile ranks can be estimated using **z-scores***:

$$z = \frac{X - \mu}{\sigma}$$

Where:

- X= score (or percentage)
- μ = mean
- σ = standard deviation

The z-score can be mapped to a **percentile rank** using a standard normal distribution table.

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4. Example

Consider a test scored out of 100. Ten students received the following scores (sorted in ascending order):

$$Scores = [55,60,62,65,68,70,72,75,78,85]$$

Suppose a student scored **70**:

- **Percentage**:

$$\frac{70}{100} \times 100 = 70\%$$

Percentile Rank:

  - L=5 scores below 70: [55, 60, 62, 65, 68]
  - E=1 score equal to 70
  - N=10 total values

$$\text{Percentile Rank} = \left(\frac{5 + 0.5(1)}{10}\right) \times 100 = \left(\frac{5.5}{10}\right) \times 100 = 55^{\text{th}} \text{ percentile}$$

- **Percentile (e.g., 70th percentile value)**:

$$\text{Index} = \frac{70}{100} \times (10 + 1) = 0.7 \times 11 = 7.7$$

Interpolating between the 7th and 8th values (i.e., 72 and 75):

$$P_{70} = 72 + 0.7 \times (75 - 72) = 72 + 2.1 = 74.1$$

So, the **70th percentile (value)** is approximately **74.1**.

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5. Conclusion

Although **percentage**, **percentile rank**, and **percentile** all relate to the concept of measurement on a scale of 100, they serve fundamentally different purposes:

- **Percentage** reflects absolute performance.
- **Percentile rank** indicates relative standing within a group.
- **Percentile** identifies the actual value that separates a certain proportion of the dataset.

Understanding these distinctions is essential in statistical reporting, educational assessments, and data analysis.